

矩阵分析与应用

第十二讲 矩阵分解之一

本讲主要内容

- 三角分解
- Givens变换
- HouseHolder变换
- QR分解

[引 入] 在线性代数中应用 Gauss 消去法求解 n 元

线性方程组 $\mathbf{Ax} = \mathbf{b}$

其中: $A = (a_{ij})_{n \times n}$, $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, $\mathbf{b} = (b_1, b_2, \dots, b_n)^T$

Gauss 消去法将系数矩阵化为上三角形矩阵, 或将增广矩阵化为上阶梯形矩阵, 而后回代求解。

$$\begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ & & \ddots & \vdots \\ & & & a_{nn}^{(n)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(n)} \end{bmatrix}$$
$$\begin{aligned} x_n &= b_n^{(n)} / a_{nn}^{(n)} \\ x_i &= \left(b_i^{(i)} - \sum_{j=i+1}^n a_{ij}^{(i)} x_j \right) / a_{ii}^{(i)} \\ & \quad (i = n-1, \dots, 1) \end{aligned}$$

定义4.1 如果 n 阶矩阵 A 能够分解为一个下三角矩阵 L 和一个上三角矩阵 U 的乘积，则称其为三角分解或 LU 分解。如果方阵 A 可分解成 $A=LDU$ ，其中 L 为一个单位下三角矩阵， D 为对角矩阵，则称 A 可作 LDU 分解。

定理4.1 矩阵 $A = (a_{ij})_{n \times n}$ 的分解式唯一的充要条件为 A 的顺序主子式 $\Delta_k \neq 0$ 。 $A=LDU$ ， 其中 L 是单位下三角矩阵， U 是单位上三角矩阵， D 是对角矩阵，并且

$$D = \text{diag}(d_1, d_2, \dots, d_n), d_k = \frac{\Delta_k}{\Delta_{k-1}}, k = 1, 2, \dots, n \quad (\Delta_0 = 1)$$

推论 设 A 是 n 阶非奇异矩阵, A 有三角分解 $A=LU$,
的充要条件是 A 的顺序主子式 $\Delta_k \neq 0 \quad k=1,2,\cdots,n$

分解原理：以 $n=4$ 为例

$$\Delta_1(A) = a_{11} : a_{11} \neq 0 \Rightarrow c_{i1} = \frac{a_{i1}}{a_{11}} \quad (i = 2, 3, 4)$$

$$L_1 = \begin{bmatrix} 1 & & & \\ c_{21} & 1 & & \\ c_{31} & 0 & 1 & \\ c_{41} & 0 & 0 & 1 \end{bmatrix}, L_1^{-1} = \begin{bmatrix} 1 & & & \\ -c_{21} & 1 & & \\ -c_{31} & 0 & 1 & \\ -c_{41} & 0 & 0 & 1 \end{bmatrix}$$

$$L_1^{-1}A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ a_{32}^{(1)} & a_{33}^{(1)} & a_{34}^{(1)} \\ a_{42}^{(1)} & a_{43}^{(1)} & a_{44}^{(1)} \end{bmatrix} = A^{(1)}$$

$$(2) \quad \Delta_2(A) = \Delta_2(A^{(1)}) = a_{11}a_{22}^{(1)} : a_{22}^{(1)} \neq 0$$

$$\Rightarrow c_{i2} = \frac{a_{i2}^{(1)}}{a_{22}^{(1)}} \quad (i = 3, 4)$$

$$L_2 = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & c_{32} & 1 & \\ 0 & c_{42} & 0 & 1 \end{bmatrix}, L_2^{-1} = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & -c_{32} & 1 & \\ 0 & -c_{42} & 0 & 1 \end{bmatrix}$$

$$L_2^{-1} A^{(1)} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ & & a_{33}^{(2)} & a_{34}^{(2)} \\ & & a_{43}^{(2)} & a_{44}^{(2)} \end{bmatrix} = A^{(2)}$$

$$(3) \Delta_3(A) = \Delta_3(A^{(2)}) = a_{11}a_{22}^{(1)}a_{33}^{(2)} : a_{33}^{(2)} \neq 0$$

$$\Rightarrow c_{43} = \frac{a_{43}^{(2)}}{a_{33}^{(2)}}$$

$$L_3 = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & c_{43} & 1 \end{bmatrix}, L_3^{-1} = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & -c_{43} & 1 \end{bmatrix}$$

$$L_3^{-1}A^{(2)} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ & & a_{33}^{(2)} & a_{34}^{(2)} \\ & & & a_{44}^{(3)} \end{bmatrix} = A^{(3)}$$

即: $L_3^{-1}L_2^{-1}L_1^{-1}A = A^{(3)} \Rightarrow A = L_1L_2L_3A^{(3)}$

令 $L = L_1L_2L_3 = \begin{bmatrix} 1 & & & \\ c_{21} & 1 & & \\ c_{31} & c_{32} & 1 & \\ c_{41} & c_{42} & c_{43} & 1 \end{bmatrix}$ 则 $A = LA^{(3)}$

分解 $A^{(3)} = \begin{bmatrix} a_{11} & & & \\ & a_{22}^{(1)} & & \\ & & a_{33}^{(2)} & \\ & & & a_{44}^{(3)} \end{bmatrix} \begin{bmatrix} 1 & * & * & * \\ & 1 & * & * \\ & & 1 & * \\ & & & 1 \end{bmatrix} = DU$

则 $A = LDU$

二、紧凑格式算法: $A = LDU = \tilde{L}U$ (Crout分解)

$$L = \tilde{L} = \begin{bmatrix} l_{11} & & & \\ l_{21} & l_{22} & & \\ \vdots & \vdots & \ddots & \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix}, U = \begin{bmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & 1 & \cdots & u_{2n} \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}$$

二、紧凑格式算法: $A = LDU = \tilde{L}U$ (Crout分解)

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & & & \\ l_{21} & l_{22} & & \\ \vdots & \vdots & \ddots & \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & 1 & & u_{2n} \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}$$

$$(i,1)\text{元: } a_{i1} = l_{i1} \bullet 1 \Rightarrow l_{i1} = a_{i1} \quad (i = 1, \cdots, n)$$

$$(1,j)\text{元: } a_{1j} = l_{11} \bullet u_{1j} \Rightarrow u_{1j} = \frac{a_{1j}}{l_{11}} \quad (j = 2, \cdots, n)$$

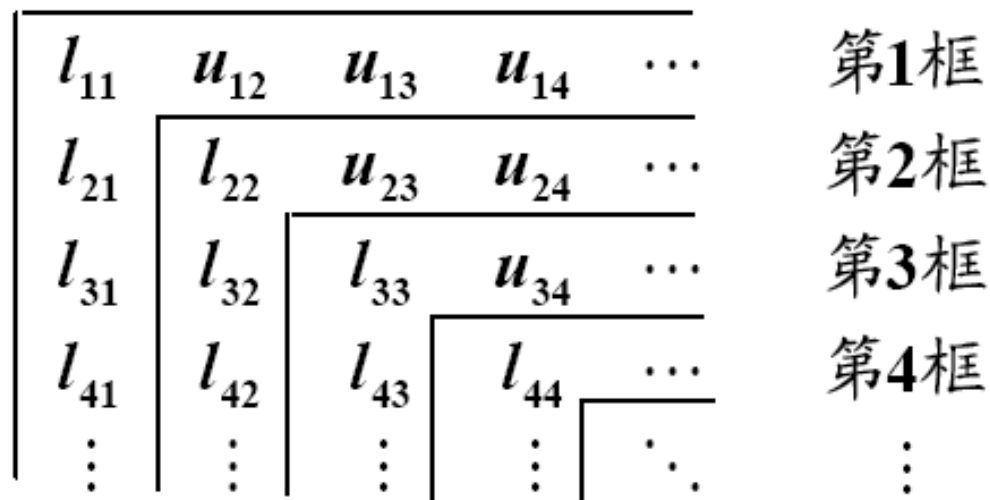
$$(i,k)\text{元: } a_{ik} = l_{i1} \bullet u_{1k} + \cdots + l_{i,k-1} \bullet u_{k-1,k} + l_{ik} \bullet 1 \quad (i \geq k)$$

$$\Rightarrow l_{ik} = a_{ik} - (l_{i1} \bullet u_{1k} + \cdots + l_{i,k-1} \bullet u_{k-1,k})$$

$$(k,j)\text{元: } a_{kj} = l_{k1} \bullet u_{1j} + \cdots + l_{k,k-1} \bullet u_{k-1,j} + l_{kk} \bullet u_{kj} \quad (j > k)$$

$$\Rightarrow u_{kj} = \frac{1}{l_{kk}} \left[a_{kj} - (l_{k1} \bullet u_{1j} + \cdots + l_{k,k-1} \bullet u_{k-1,j}) \right]$$

计算框图：



$$l_{i1} = a_{i1} \quad (i = 1, \cdots, n)$$

$$u_{1j} = \frac{a_{1j}}{l_{11}} \quad (j = 2, \cdots, n)$$

$$l_{ik} = a_{ik} - \left(l_{i1} \bullet u_{1k} + \cdots + l_{i,k-1} \bullet u_{k-1,k} \right) \quad (i \geq k)$$

$$u_{kj} = \frac{1}{l_{kk}} \left[a_{kj} - \left(l_{k1} \bullet u_{1j} + \cdots + l_{k,k-1} \bullet u_{k-1,j} \right) \right] \quad (j \leq k)$$

$$A = \begin{bmatrix} 5 & 2 & -4 & 0 \\ 2 & 1 & -2 & 1 \\ -4 & -2 & 5 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

计算框图:

$$\begin{array}{c|c|c|c} 5 & 2/5 & -4/5 & 0 \\ \hline 2 & 1/5 & -2 & 5 \\ \hline -4 & -2/5 & 1 & 2 \\ \hline 0 & 1 & 2 & -7 \end{array}$$

$$l_{i1} = a_{i1} \quad (i = 1, \dots, n)$$

$$u_{1j} = \frac{a_{1j}}{l_{11}} \quad (j = 2, \dots, n)$$

$$l_{ik} = a_{ik} - (l_{i1} \bullet u_{1k} + \dots + l_{i,k-1} \bullet u_{k-1,k}) \quad (i \geq k)$$

$$u_{kj} = \frac{1}{l_{kk}} \left[a_{kj} - (l_{k1} \bullet u_{1j} + \dots + l_{k,k-1} \bullet u_{k-1,j}) \right] \quad (j \leq k)$$

$$A = \begin{bmatrix} 5 & 2 & -4 & 0 \\ 2 & 1 & -2 & 1 \\ -4 & -2 & 5 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

计算框图:

$$\begin{array}{c|c|c|c} 5 & 2/5 & -4/5 & 0 \\ \hline 2 & 1/5 & -2 & 5 \\ \hline -4 & -2/5 & 1 & 2 \\ \hline 0 & 1 & 2 & -7 \end{array}$$

$$\tilde{L} = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 2 & 1/5 & 0 & 0 \\ -4 & -2/5 & 1 & 0 \\ 0 & 1 & 2 & -7 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ 2/5 & 1 & & \\ -4/5 & -2 & 1 & \\ 0 & 5 & 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & & & \\ & 1/5 & & \\ & & 1 & \\ & & & -7 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 2/5 & -4/5 & 0 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A = \tilde{L}U = LDU$$

QR分解

目的：将分解为正交矩阵与上三角矩阵之积.

约定：本节涉及的矩阵为实矩阵，向量为实向量，数为实数.

一、Givens矩阵

$$T_{ij}(c, s) \triangleq \begin{bmatrix} I & & & \\ & c & & s \\ & & I & \\ & -s & & c \\ & & & & I \end{bmatrix} \begin{matrix} (i) \\ \\ (j) \end{matrix} \quad c^2 + s^2 = 1$$

性质:

$$(1) \quad T_{ij}^T T_{ij} = I, [T_{ij}(c, s)]^{-1} = [T_{ij}(c, s)]^T = T_{ij}(c, -s), \quad |T_{ij}| = 1$$

$$(2) \quad x = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}, \quad T_{ij} x \triangleq \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_n \end{bmatrix} \Rightarrow \begin{cases} \eta_i = c\xi_i + s\xi_j \\ \eta_j = -s\xi_i + c\xi_j \\ \eta_k = \xi_k \quad (k \neq i, j) \end{cases}$$

$$\text{若 } \xi_i^2 + \xi_j^2 \neq 0, \text{ , 取 } c = \frac{\xi_i}{\sqrt{\xi_i^2 + \xi_j^2}}, s = \frac{\xi_j}{\sqrt{\xi_i^2 + \xi_j^2}}$$

$$\text{则 } \eta_i = \sqrt{\xi_i^2 + \xi_j^2} > 0, \quad \eta_j = 0$$

定理3: $\mathbf{x} \neq \mathbf{0} \Rightarrow \exists$ 有限个 **G**-矩阵之积 $\mathbf{T}$, st. $\mathbf{T}\mathbf{x} = |\mathbf{x}|\mathbf{e}_1$

推论: 设非零列向量 $\mathbf{x} \in \mathbf{R}^n$ 及单位列向量 $\mathbf{z} \in \mathbf{R}^n$,

则存在有限个Givens矩阵之积, 记作 $\mathbf{T}$, 使得

$$\mathbf{T}\mathbf{x} = |\mathbf{x}|\mathbf{z}$$

例: $x = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ 求G-矩阵之积 T ,使得 $Tx = |x|e_1$

解: $T_{12}(c,s)$ 中, $c = \frac{3}{5}, s = \frac{4}{5}$. $T_{12}x = \begin{bmatrix} 5 \\ 0 \\ 5 \end{bmatrix}$

$T_{13}(c,s)$ 中, $c = \frac{1}{\sqrt{2}}, s = \frac{1}{\sqrt{2}}$. $T_{13}(T_{12}x) = \begin{bmatrix} 5\sqrt{2} \\ 0 \\ 0 \end{bmatrix} = |x|e_1$

$$T = T_{13}T_{12} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ -1 & 0 & 1 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 & 4 & 0 \\ -4 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \frac{1}{5\sqrt{2}} \begin{bmatrix} 3 & 4 & 5 \\ -4\sqrt{2} & 3\sqrt{2} & 0 \\ -3 & -4 & 5 \end{bmatrix}$$

$$Tx = 5\sqrt{2}e_1$$

Householder矩阵

在平面 $\mathbf{R}^2$ 中，将向量 x 映射为关于 e_1 对称的向量 y 的变换，称为是关于 e_1 轴的镜像(反射)变换

设 $x = \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}$ ，有

$$y = \begin{bmatrix} \xi_1 \\ -\xi_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = (I - 2e_2e_2^T)x = Hx$$

其中， $e_2 = [0 \ 1]^T$ ， H 是正交矩阵，且 $|H| = -1$

Householder矩阵

将向量 x 映射为关于“与单位向量 u 正交的直线”

对称的向量 y 的变换, $x - y = 2u(u^T x)$

$$y = x - 2u(u^T x) = (I - 2uu^T)x = Hx$$

显然, H 是正交矩阵

$$\begin{aligned} H^T H &= (I - 2uu^T)^T (I - 2uu^T) \\ &= (I - 2uu^T)(I - 2uu^T) \\ &= I + 4\mathbf{u}\mathbf{u}^T\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{u}^T \\ &= I + 4\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{u}^T = I \end{aligned}$$

Householder矩阵

将向量 x 映射为关于“与单位向量 u 正交的直线”

对称的向量 y 的变换, $x - y = 2u(u^T x)$

$$y = x - 2u(u^T x) = (I - 2uu^T)x = Hx$$

显然, H 是正交矩阵

定义: 设单位列向量 $u \in R^n$, 称 $H = I - 2uu^T$

为Householder矩阵(初等反射矩阵), 由 H 矩阵确定的线性变换称为Householder变换。

Householder矩阵

$$H_u = I_n - 2uu^T \quad (u \in R^n \text{ 是单位列向量})$$

$$(1) H = H^T \text{ 对称}$$

$$(2) H^T H = I \text{ 正交}$$

$$(3) H^2 = I \text{ 对合}$$

$$(4) H^{-1} = H \text{ 自逆}$$

$$(5) \det H = -1 \text{ 自逆}$$

验证(5):

$$\begin{bmatrix} I & 0 \\ -u^T & 1 \end{bmatrix} \begin{bmatrix} I & 2u \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} I - 2uu^T & 0 \\ u^T & 1 \end{bmatrix} = \begin{bmatrix} I & 0 \\ -u^T & 1 \end{bmatrix} \begin{bmatrix} I & 2u \\ u^T & 1 \end{bmatrix} = \begin{bmatrix} I & 2u \\ 0^T & -1 \end{bmatrix}$$

$$\begin{vmatrix} I - 2uu^T & 0 \\ u^T & 1 \end{vmatrix} = \begin{vmatrix} I & 2u \\ 0^T & -1 \end{vmatrix} = -1$$

定理4: R^n 中 $(n > 1)$, $\forall x \neq 0, \forall$ 单位列向量 z

$$\Rightarrow \exists H_u, \text{st } H_u x = |x|z$$

证明: (1) $x = |x|z$: $n > 1$ 时, 取单位向量 u 使得 $u \perp x$,

于是 $H_u = I - 2uu^T$: $H_u x = Ix - 2uu^T x = x = |x|z$

(2) $x \neq |x|z$: 取 $u = \frac{x - |x|z}{|x - |x|z|}$, 有

$$H_u x = \left[I - 2 \frac{(x - |x|z)(x - |x|z)^T}{|x - |x|z|^2} \right] x = x - \frac{2(x - |x|z, x)}{|x - |x|z|^2} (x - |x|z)$$

$$(x - |x|z, x) = |x|^2 - |x|(z, x) \quad |x - |x|z|^2 = (x - |x|z, x - |x|z) = 2|x|^2 - 2|x|(z, x)$$

定理4: R^n 中 $(n > 1), \forall x \neq 0, \forall$ 单位列向量 z

$$\Rightarrow \exists H_u, \text{st } H_u x = |x|z$$

证明: (1) $x = |x|z: n > 1$ 时, 取单位向量 u 使得 $u \perp x$,

于是 $H_u = I - 2uu^T: H_u x = Ix - 2uu^T x = x = |x|z$

(2) $x \neq |x|z$: 取 $u = \frac{x - |x|z}{|x - |x|z|}$, 有

$$\begin{aligned} H_u x &= \left[I - 2 \frac{(x - |x|z)(x - |x|z)^T}{|x - |x|z|^2} \right] x = x - \frac{2(x - |x|z, x)}{|x - |x|z|^2} (x - |x|z) \\ &= x - 1 \times (x - |x|z) = |x|z \end{aligned}$$

例2: $x = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, 求H-矩阵 H 使得 $Hx = |x|e_1$

解: $|x| = 3, x - |x|e_1 = \begin{bmatrix} -2 \\ 2 \\ 2 \end{bmatrix}, \quad u = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$$H = I - \frac{2}{3} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

$$Hx = 3e_1$$

G矩阵与H-矩阵的关系

定理5: G-矩阵 $T_{ij}(c, s) \Rightarrow \exists$ H-矩阵 H_u 与 H_v , st $T_{ij} = H_u H_v$

证明: $c^2 + s^2 = 1 \Rightarrow$ 取 $\theta = \arctan \frac{s}{c}$, 则 $\cos \theta = c, \sin \theta = s$

$$T_{ij}(c, s) = \begin{bmatrix} I & & & \\ & \cos \theta & \sin \theta & \\ & & I & \\ & -\sin \theta & \cos \theta & \\ & & & I \end{bmatrix} \begin{matrix} (i) \\ \\ (j) \\ \end{matrix}$$

$$v = \begin{bmatrix} 0 & \dots & 0 & \sin \frac{\theta}{4} & 0 & \dots & 0 & \cos \frac{\theta}{4} & 0 & \dots & 0 \end{bmatrix}^T$$

$$\begin{aligned}
 H_v &= \begin{bmatrix} \textcolor{red}{I} & & & \\ & 1 & & \\ & & \textcolor{red}{I} & \\ & & & 1 \\ & & & & \textcolor{red}{I} \end{bmatrix} - 2 \begin{bmatrix} \textcolor{red}{O} & \sin^2 \frac{\theta}{4} & \sin \frac{\theta}{4} \cos \frac{\theta}{4} & & \\ & & \textcolor{red}{O} & & \\ \sin \frac{\theta}{4} \cos \frac{\theta}{4} & & & \cos^2 \frac{\theta}{4} & \\ & & & & \textcolor{red}{O} \\ & & & & & \textcolor{red}{O} \end{bmatrix} \\
 &= \begin{bmatrix} \textcolor{red}{I} & & & & \\ & \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} & & \\ & & \textcolor{red}{I} & & \\ & -\sin \frac{\theta}{2} & -\cos \frac{\theta}{2} & & \\ & & & & \textcolor{red}{I} \end{bmatrix}
 \end{aligned}$$

$$u = \left[0 \ \cdots \ 0 \ \sin \frac{3\theta}{4} \ 0 \ \cdots \ 0 \ \cos \frac{3\theta}{4} \ 0 \ \cdots \ 0 \right]^T$$

$$H_u = \begin{bmatrix} I & & & \\ & \cos \frac{3\theta}{2} & -\sin \frac{3\theta}{2} & \\ & & I & \\ & -\sin \frac{3\theta}{2} & -\cos \frac{3\theta}{2} & \\ & & & I \end{bmatrix},$$

$$T_{ij}(c, s) = H_u H_v \quad \#$$

[注] H-矩阵不能由若干个G矩阵的乘积来表示。

因为 $\det H = -1$, 而 $\det G = 1$

例3: G-矩阵 $T_{ij}(0,1) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ 中, $c = 0, s = 1 \Rightarrow \theta = \pi/2$

$$H_u = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}, H_v = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$\Rightarrow H_u H_v = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

四、QR分解

1. Schmidt正交化方法

定理6: $A_{n \times n}$ 可逆 $\Rightarrow \exists$ 正交矩阵 Q , 可逆上三角矩阵 R , 使得 $A=QR$ 。

证明: $A=(a_1, a_2, \dots, a_n)$ 可逆 $\Rightarrow a_1, a_2, \dots, a_n$ 线性无关,
正交化后可得:

$$\begin{cases} b_1 = a_1 \\ b_2 = a_2 - k_{21}b_1 \\ \dots\dots\dots \\ b_n = a_n - k_{n,n-1}b_{n-1} - \dots - k_{n1}b_1 \end{cases} \quad \begin{cases} a_1 = b_1 \\ a_2 = k_{21}b_1 + b_2 \\ \dots\dots\dots \\ a_n = k_{n1}b_1 + \dots + k_{n,n-1}b_{n-1} + b_n \end{cases}$$

$$(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)K$$

$$\begin{cases} a_1 = b_1 \\ a_2 = k_{21}b_1 + b_2 \\ \dots\dots\dots \\ a_n = k_{n1}b_1 + \dots + k_{n,n-1}b_{n-1} + b_n \end{cases}$$

$$= (q_1, q_2, \dots, q_n) \begin{bmatrix} |b_1| & & & \\ & |b_2| & & \\ & & \ddots & \\ & & & |b_n| \end{bmatrix} \begin{bmatrix} 1 & k_{21} & \dots & k_{n1} \\ & 1 & \dots & k_{n2} \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}$$

$$\text{令 } Q = (q_1, q_2, \dots, q_n), R = \begin{bmatrix} |b_1| & & & \\ & |b_2| & & \\ & & \ddots & \\ & & & |b_n| \end{bmatrix} \begin{bmatrix} 1 & k_{21} & \dots & k_{n1} \\ & 1 & \dots & k_{n2} \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}$$

$$\text{则 } A=QR, \text{ 其中 } q_i = \frac{b_i}{|b_i|} \quad (i = 1, 2, \dots, n)$$

例4: 求 $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ 的QR分解。

$$\text{解: } b_1 = a_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, b_2 = a_2 - 1 \times b_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, b_3 = a_3 - \frac{1}{3}b_2 - \frac{7}{6}b_1 = \begin{bmatrix} 1/2 \\ 0 \\ -1/2 \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad R = \begin{bmatrix} \sqrt{6} & & \\ & \sqrt{3} & \\ & & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 1 & \frac{7}{6} \\ & 1 & \frac{1}{3} \\ & & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{6} & \sqrt{6} & \frac{7}{\sqrt{6}} \\ & \sqrt{3} & \frac{1}{\sqrt{3}} \\ & & \frac{1}{\sqrt{2}} \end{bmatrix}$$

定理7: $A_{m \times n}$ 列满秩 $\Rightarrow \exists$ 矩阵 $Q_{m \times n}$ 满足 $Q^H Q = I$,
可逆上三角矩阵 $R_{n \times n}$, 使得 $A = QR$ 。

证明：同定理6